

# QUANTUMCROP AI – ANALYSIS REPORT

Quantum Remote Sensing & Yield Prediction Platform

Date: 2026-07-26

Sensor: Simulated Sentinel-2 Multispectral

Analysis Engine: Classical + QML (Hybrid)

Location ID: Field-Alpha-32

## Satellite & Index Imagery



Uploaded Imagery (RGB)

[NDVI Index Map Not Generated]

Computed NDVI Index Map

## Crop Health & Yield Prediction Summary

|                        |                     |
|------------------------|---------------------|
| Crop Health Status:    | PermanentCrop       |
| Prediction Confidence: | 52.60%              |
| Estimated Crop Yield:  | 6.38 Tons / Hectare |

## Machine Learning Benchmarking (Classical vs Quantum)

| Model Name    | Type           | Health Classification | Confidence | Accuracy (Benchmark) | Inference Latency |
|---------------|----------------|-----------------------|------------|----------------------|-------------------|
| RANDOM FOREST | Classical      | PermanentCrop         | 42.0%      | 88.0%                | ~0.002s           |
| SVM           | Classical      | PermanentCrop         | 52.6%      | 86.0%                | ~0.002s           |
| QSVM          | Quantum (QSVM) | PermanentCrop         | 43.3%      | 84.0%                | ~0.005s           |

## AI Assistant Recommendations

- **[HIGH] Low Vegetation:** Warning: Very low vegetation density. Indicates potential crop stress or sparse growth.
- **[HIGH] Nitrogen Deficiency:** Extremely low Chlorophyll Index detected. Apply urea or targeted nitrogen fertilizers within 5 days.
- **[MEDIUM] Data Quality Warning:** Prediction confidence is low. Recommend collecting more high-resolution drone imagery.